

Sensorless Position Measurement and Blocking Detection Based on Back EMF sensing for Stepper Motor in Next-generation Spectrograph Robots

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ABSTRACT

Miniature stepper motors in high-precision robotic fiber positioners require sensorless blocking detection. This paper proposes a back Electromotive Force (EMF) sensing method. By splitting the motor phases and dedicating half of the coils for measurement, they achieve reliable stall detection and position measurement with minimal analog electronics.

RELEVANCE TO THE EB15 ROBOTIC ARM PROJECT

Provides theoretical support for detecting mechanical blockage and stall without external physical encoders, serving as a backup security mechanism for the EB15 joints.

TECHNICAL ANALYSIS & SYSTEM INTEGRATION

This academic research reference documents crucial design paradigms directly relevant to the development of the EB15 six-degrees-of-freedom robotic manipulator. Under the distributed control framework designed for this thesis (featuring the ESP32 S3 as master and the Arduino Uno as slave), the mathematical formulations, communication latencies, and performance profiles analyzed by Laurent Jenni provide a benchmark for physical and algorithmic modeling. Specifically, this reference establishes solid validation criteria for timing constraints, closed-loop feedback via absolute magnetic encoders (AS5600), and vibration damping.